



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B. Tech (CBCS) Degree Examinations
May 2024
RI2005-1 – INTRODUCTION TO OBJECT ORIENTED PROGRAMMING
(RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What data type is the object in `L = [1, 23, 'hello', 1]`
A) List B) Dictionary
C) Array D) Tuple
2. Which of these is not a core data type?
A) List B) Dictionary
C) class D) Tuple
3. Which type of loop can be used to perform the following iteration: You choose a positive integer at random and then print the numbers from 1 up to and including the selected integer.
A) a for-loop or a while-loop B) only a while-loop
C) only a for-loop D) None of the above
4. Which of the following statements will not be printed when this Python code is run?
for letter in 'Python':
 if letter == 'h':
 continue
 print('Current Letter : ' + letter)
A) Current Letter: P B) Current Letter: h
C) Current Letter: t D) Current Letter: o
5. What will the following code print?
for i in range(1,4):
 for j in range(1,4):
 print(i, j, end=' ')
A) 1 1 2 2 3 3 B) 1 2 3 1 2 3 1 2 3
C) 1 1 2 1 3 1 2 1 2 2 2 3 3 1 3 2 3 3 D) 1 1 1 2 1 3 2 1 2 2 2 3 3 1 3 2 3 3
6. How dictionary is defined in Python?
A) Using curly braces {} B) Using angle brackets <>
C) Using parentheses () D) Using square brackets []
7. What is the output of the following Python code: `type(8)`?
A) <class 'int'> B) <class 'integer'>
C) <class 'number'> D) <class 'digit'>
8. In which of the following does the CricketFan class correctly inherit from the Person class?
A) from person import cricket B) pr = Cricketfan()
C) class CricketFan(Person) D) CricketFan = Person()
9. Which of the following does not correctly create an object instance?
A) puppy = Dog("Jamie") B) dog = Dog("Jamie")
C) jamie = Dog() D) pupper = new Dog("Jamie")
10. _____ is not a keyword, but by convention it is used to refer to the current instance (object) of a class.
A) class B) self
C) def D) init
11. All classes have a function called?
A) __init__ B) init
C) __init__() D) init()

12. A variable that is defined inside a method and belongs only to the current instance of a class is known as?
 - A) Inheritance
 - B) Function overloading
 - C) Instance variable
 - D) Instantiation
13. A class variable or instance variable that holds data associated with a class and its object is known as?
 - A) Class variable
 - B) Operator overloading
 - C) Method
 - D) Data member
14. _____ represents an entity in the real world with its identity and behavior.
 - A) A method
 - B) A class
 - C) An object
 - D) An operator
15. Who developed Python Tkinter?
 - A) Armin Ronacher
 - B) John Ousterhout
 - C) Travis Oliphant
 - D) Rasmus Lerdorf
16. Which widget are used to get the data from the user?
 - A) Entry
 - B) Button
 - C) Label
 - D) None of these
17. Essential thing to create a window screen using tkinter python?
 - A) create a button
 - B) call Tk() function
 - C) To define a geometry
 - D) All of these
18. Which of the following we will be able to delete using destroy() function ?
 - A) Button
 - B) Label
 - C) Frame
 - D) All of these
19. Which of the following tool provides a GUI in python?
 - A) Numpy
 - B) Tkinter
 - C) Opencv
 - D) Scipy
20. What is widget in Tkinter GUI in python
 - A) That will work in background
 - B) That will display on the screen
 - C) Change the font size
 - D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain scope of a variable in python programming using suitable programming examples.	5	L*2	1	1,2
	b) Develop a python program to implement list operations.	5	L2	1	1,2
	c) Define debugging. Explain how to debug the program using pdb module and breakpoint() with program example.	6	L2	2	1,2
2.	a) What is type casting? Explain types of type conversion with program example.	5	L2	1	1,2
	b) What are generators? Develop a python program to find factorial of a number using generators.	5	L2	1	1,2
	c) Explain with suitable example of iterables and iterators in python.	6	L2	2	1,2
3.	a) Explain break and continue in python with suitable program example.	5	L2	1	1,2
	b) Explain any 5 methods used in python dictionary with examples for each.	5	L2	1	1,2
	c) Explain types of errors in python with example. Demonstrate how to handle exceptions with program example.	6	L2	2	1,2
Unit – II					
4.	a) What are different types of inheritance in python? Explain with suitable example.	8	L2	3	1,2
	b) Explain method overriding with suitable program example in python.	8	L2	4	1,2
5.	a) Explain object and class with syntax and example in python.	8	L2	3	1,2
	b) Define modules. Develop a python program to implement rock, paper and scissors game by importing a random module in python.	8	L2	4	1,2

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| 6. | a) | What is virtual functions in python? Explain virtual functions with example. | 8 | L2 | 3 | 1,2 |
| | b) | Define Package. Write a python program to create a package(college), sub package (all dept), modules (departments) and create an admin and cabin function to the module. | 8 | L2 | 4 | 1,2 |

Unit – III

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|----|----|---|---|----|---|-----|
| 7. | a) | What is Tkinter? Explain the use cases of Tkinter. | 8 | L2 | 5 | 1,2 |
| | b) | Develop a python program to demonstrate event driven programming and explain. | 8 | L2 | 5 | 1,2 |
| 8. | a) | Develop a python program to add buttons with Tkinter and explain. | 8 | L2 | 5 | 1,2 |
| | b) | Explain three major geometry managers used to control layouts. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
